Case 2: Day-ahead forecasting of national electricity consumption

Based on Prosçanu, Prosçanu, Pele & Costea (2025), *Watt's next? Benchmarking time series models on Romania's national electricity consumption*, Romanian Journal of Economic Forecasting 28(3), 146–163, RePEc

The business question

Every morning, a Romanian supplier or the balancing team of a producer must notify the transmission system operator (TSO), Transelectrica, of the volume it will consume or deliver in each hour of the next day. Every MWh of error becomes an *imbalance*, settled at a price that is usually worse than the day-ahead price. The head of trading must decide: **which forecasting model do we run in production — a classical seasonal model, a machine-learning model, or a pretrained "foundation" model — and is the gain worth the cost?**

1 What the paper did

Question. The paper benchmarks univariate forecasting models on Romania's national electricity consumption, (1) on predictive accuracy (MSE, MAE, R^2) and (2) on the effect of model architecture (p. 147).

Data. Hourly national consumption, 1 January 2022 – 31 December 2024, 26,304 observations in MWh (sistemulenergetic.ro); mean 6,230.59, minimum 2,871.75, maximum 9,210.75 (Table 1, p. 149). Lowest demand at 2–5 a.m., peak around 7 p.m.; winter months highest (p. 150).

Method. Nine models: SARIMA (Seasonal AutoRegressive Integrated Moving Average), the neural networks N-BEATS, NHITS, LSTM (Long Short-Term Memory), Mamba, Titans and RetNet, and two pretrained foundation models used zero-shot, Chronos-T5 and TimesFM. Look-back 168 hours, horizon 24 hours, chronological 80/20 split (Table 3, p. 157).

Main findings of the paper.

- "The N-BEATS model has been the most successful", with MAPE 2.324 % and "the highest coefficient of determination ($R^2 = 0.956$)" (p. 158; Table 4, p. 157). N-BEATS "lowers MAPE from 3.63 % (SARIMA) to 2.32 %" (p. 159).
- RetNet and Titans also perform very well (p. 158–159).
- "TimesFM consistently underperforms" (MAPE 5.895 %, $R^2 = 0.673$); Chronos-T5 is the fastest (7 seconds) with MAPE 3.325 % (p. 159; Table 4).
- SARIMA "remains a viable and efficient option" when resources are constrained; neural models suit capacity planning and renewable integration (p. 160).

2 The method, step by step

Intuition first

Consumption repeats itself: every day has the same shape, every week has quiet weekends, every year has cold winters. A forecasting model is a machine for copying the right past pattern and adjusting it. Simple models copy one pattern ("two days ago", "last week"); better models learn which past hours matter and how much.

Let $L_{d,h}$ be the load (MW) in hour h of day d. At gate closure, 12:00 CET on day $d-1$, we forecast the 24 values $\hat{L}_{d,0}, \ldots, \hat{L}_{d,23}$. At that time load is metered only up to the end of day $d-2$, so:

- ⊡ every model uses load data up to day $d-2$ only (lags of at least 48 hours, plus $d-7$);
- ⊡ no value of day $d-1$ is used, not even its daily mean.

The *naive* forecasts are $\hat{L}_{d,h} = L_{d-2,h}$ (two days earlier) and $\hat{L}_{d,h} = L_{d-7,h}$ (last week). SARIMA$(1,0,1)(1,1,1)_{24}$ models the 24-hour difference $L_t - L_{t-24}$ as depending on its own past and on past errors. The *linear regression* and *gradient boosting* (an ensemble of small decision trees, each fitted to the errors of the previous ones) use lagged load (48, 72, 168, 336 hours), the mean of day $d-2$, hour, weekday, month and holidays. The *neural network* maps the 168 hours up to the end of $d-2$ directly to the 24 hours of day d. Chronos-Bolt is a model pretrained on many public time series; it reads the 512 hours up to the end of $d-2$ and outputs forecast quantiles without being trained on our data (zero-shot). Accuracy is measured on all N test hours with

$$\text{MAE} = \frac{1}{N}\sum_t |L_t - \hat{L}_t|, \qquad \text{MAPE} = \frac{100}{N}\sum_t \left|\frac{L_t - \hat{L}_t}{L_t}\right|, \qquad R^2 = 1 - \frac{\sum_t (L_t - \hat{L}_t)^2}{\sum_t (L_t - \bar{L})^2}.$$

In practice

The model choice is a money question: annual imbalance volume ≈ MAE × 8,760 hours × your market share; multiply by the imbalance cost per MWh. A model is worth deploying if this saving exceeds its running and maintenance cost.

3 Replicate it yourself

Notebook `C2_notebook_EN.ipynb`: hourly load from ENTSO-E via energy-charts.info (`generation_ro_mw_hourly_utc.csv`), the paper's sample and an 80/20 split; the test period starts on 27 May 2024 (217 forecast days). Our series reproduces the paper's Table 1: 26,304 hours, mean 6,230.90 MW, minimum 2,871.75, maximum 9,210.75 — the same Transelectrica metering, published twice.

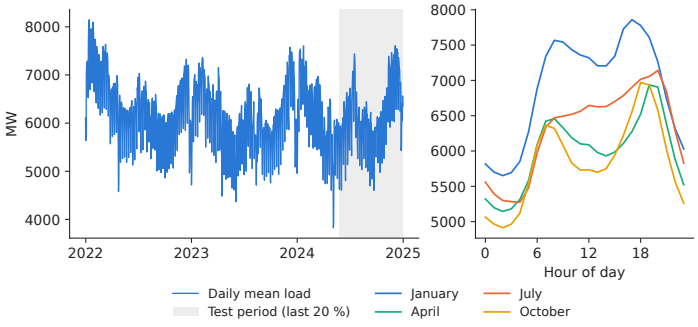


Figure 1: Daily mean load and the test period (left); mean hourly profile by month (right).

Source: energy-charts.info (ENTSO-E); 1.1.2022–31.12.2024.

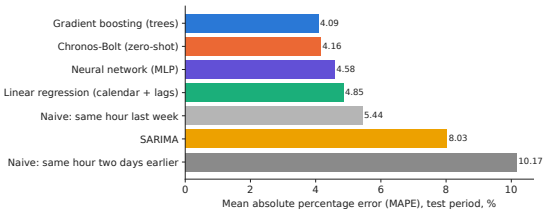


Figure 2: Test MAPE by model.

Source: our forecasts; test period 27 May 2024–31.12.2024.

Table 1: Day-ahead forecast accuracy on the test period, 217 days × 24 hours. Best value in bold (Bias closest to zero).

Model	MAE, MW	RMSE, MW	MAPE, %	R^2	Bias, MW
Gradient boosting (trees)	**250**	**339**	**4.09**	**0.879**	32
Chronos-Bolt (zero-shot)	254	360	4.16	0.863	43
Neural network (MLP)	284	393	4.58	0.837	80
Linear regression (calendar + lags)	292	383	4.85	0.845	19
Naive: same hour last week	335	474	5.44	0.762	**17**
SARIMA	483	634	8.03	0.575	34
Naive: same hour two days earlier	610	825	10.17	0.280	18

Source: our computation; Bias = mean of actual minus forecast.

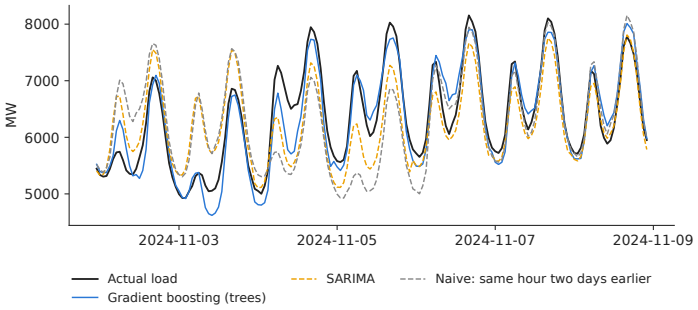


Figure 3: One week of forecasts in November 2024.

Source: our forecasts; energy-charts.info.

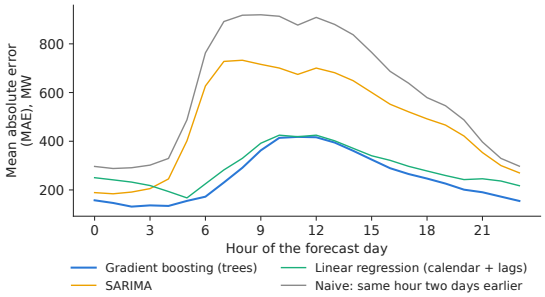


Figure 4: MAE by hour of the forecast day.

Source: our forecasts; test period.

Our result. The best model on our test period is Gradient boosting (trees) with MAPE 4.09 % (MAE 250 MW, $R^2 = 0.879$), followed closely by Chronos-Bolt used zero-shot (4.16 %). The naive "two days earlier" rule has 10.17 %, "last week" 5.44 %; our SARIMA reaches 8.03 %, with parameters estimated on 8 weeks and held fixed over the test, so it does not adapt to holidays and the summer. As in the paper, a pretrained model used zero-shot is competitive, and the ranking depends on the metric. The 10 %–90 % interval of Chronos-Bolt covered 85.1 % of the hours (target 80 %). Our numbers reflect our test window, forecast origin and model versions; the ranking is ex post (the "best" model is identified on the test period itself), so the saving below is an upper estimate. **Decision number:** for a supplier with 10 % of national consumption and an illustrative imbalance cost of 50 EUR/MWh, moving from the naive rule to the best model cuts the annual imbalance bill from 26.70 to 10.94 million EUR (saving 15.76 million EUR).

4 Discussion questions for managers

1. Using the paper's benchmarking framework, what evidence would you need before choosing a model for production: seasonal performance, imbalance cost, runtime or maintenance?
2. A pretrained model needs no training on your data. What are the risks of relying on a model you cannot inspect or retrain?
3. Which hours of the day carry the largest errors (Figure 4), and are they the expensive hours for your portfolio?
4. National load is smooth; your own portfolio (a few hundred clients, rooftop solar) is not. How would you expect the ranking of models to change?
5. Is a MAPE improvement of one percentage point worth a data-science team? Use the decision number above and your own share and imbalance cost.

5 Limitations of the classroom replication

(i) Our models are univariate: no weather or economic drivers. (ii) One test window (late May – December 2024); rankings can change with the season. (iii) National load hides the behind-the-meter solar boom; a supplier's own portfolio is noisier. (iv) The imbalance cost of 50 EUR/MWh is an illustrative assumption, not a market statistic. (v) Our neural network and Chronos version are compact implementations chosen to run in class.

Summary of Case 2

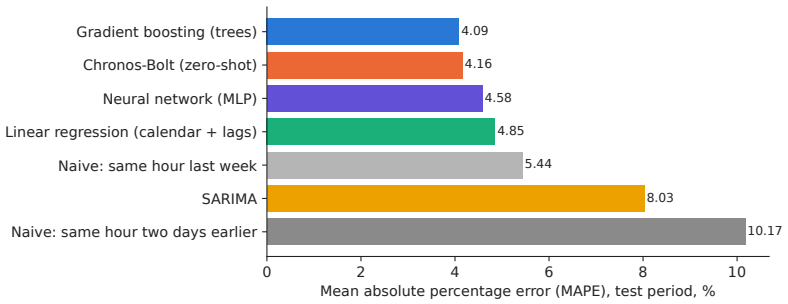
Question
Which model should forecast tomorrow's hourly consumption, and is the gain worth it?

Data
Hourly national load 2022–2024 (26,304 h), same series as the paper; last 20 % for testing.

Method
Naive, SARIMA, regression, gradient boosting, neural net, Chronos zero-shot; MAE, MAPE, R^2.

Result
Best: Gradient boosting (trees), MAPE 4.09 % vs 10.17 % naive. Paper: N-BEATS 2.324 % vs SARIMA 3.63 %.

Decision
Deploy a modern model if the imbalance saving (≈15.76 mil. EUR/yr for a 10 % supplier) exceeds its cost.



Source: our forecasts on energy-charts.info (ENTSO-E) load; test period 27 May 2024–31.12.2024.

4.09 %
best MAPE on our test

10.17 %
naive "two days earlier" rule

15.76
mil. EUR/yr imbalance saving, 10 % supplier